Syllabus

MMI5190E The New Frontiers of AI: From Perception to Agentic Intelligence

Prof. Dr. Pin Luarn

Winter Semester 2026 / 27

Level	Master	
Credits	3	
Student Contact Hours	30	
Workload	30 h	
Prerequisites	none	
Time	see LSF	
Room	see LSF	
Start Date	see LSF	
Lecturer	**Name**	Prof. Dr. Pin Luarn
	Office	-
	Virtual Office	-
	Office Hours	-
	Phone	-
	Email	luarn@mail.ntust.edu.tw, luarn@gapps.ntust.edu.tw

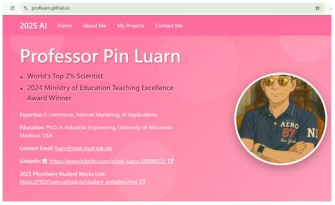
Summary

This course is designed for MBA students and focuses on the practical development of AI applications in business contexts. Students will gain a foundational understanding of AI, from Perceptive AI such as image, speech, and behavior recognition, to Generative AI for text, image, and multimodal content generation, and finally to Agentic AI, which enables autonomous agents to plan, reason, use tools, automate workflows, and execute tasks.

The course emphasizes hands-on learning with tools such as Python, MediaPipe, OpenCV, OpenAI and Google LLMs, Ollama, n8n. In addition, the course will introduce several emerging Agentic AI tools such as OpenClaw and Hermes Agent, enabling students to explore how AI agents can be designed, connected, and applied to real-world business processes.

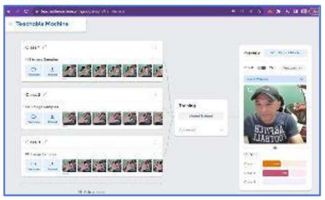
Outline of the Course

Unit 1: Introduction to AI, Development Environment Setup, and Personal Webpage
- Introduction to the core concepts of Perceptive AI, Generative AI, and Agentic AI.
- Overview of key tools: Python, IDEs, MediaPipe, OpenCV, LLM APIs, Ollama, n8n, and GitHub.
- Hands-on exercise: Setting up Python and an IDE.
- Hands-on exercise: Creating a simple personal webpage to present AI projects and course assignments.

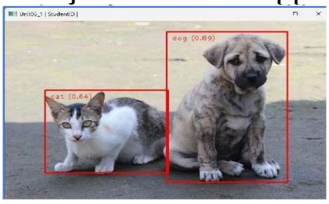




Unit 2 : Fundamentals of Visual AI: MediaPipe and Teachable Machine
- Exploring Google MediaPipe for visual AI development.
- Introduction to Google Teachable Machine for training custom image models.
- Hands-on exercise: Installing MediaPipe and OpenCV, testing basic visual AI functions, and exploring Teachable Machine.

Unit 3: Object Detection with MediaPipe
- Introduction to object detection algorithms.
- Implementing real-time object detection with MediaPipe.
- Hands-on exercise: Building a simple object detection application.

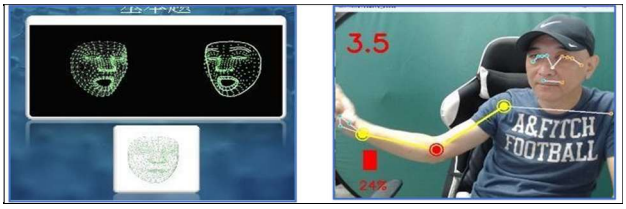


Unit 4: Image Segmentation and Face Detection with MediaPipe
- Introduction to MediaPipe's **Image Segmenter** and face detection models.
- Implementing image segmentation and real-time face detection.
- Hands-on exercise: Creating an application for face detection and image segmentation.





Unit 5: Advanced Face Landmark and Pose Recognition
- Exploring MediaPipe's **Face Landmarker** and pose estimation models.
- Implementing real-time face landmark detection and pose estimation.
- Hands-on exercise: Developing an application for face landmark recognition and human pose estimation.



Unit 6: Hand Landmark Detection & Gesture recognition
- Introduction to MediaPipe's **Hand Landmarker** and **Gesture Recognizer**.
- Implementing real-time hand landmark and gesture recognition.
- Hands-on exercise: Building a hand gesture recognition application.



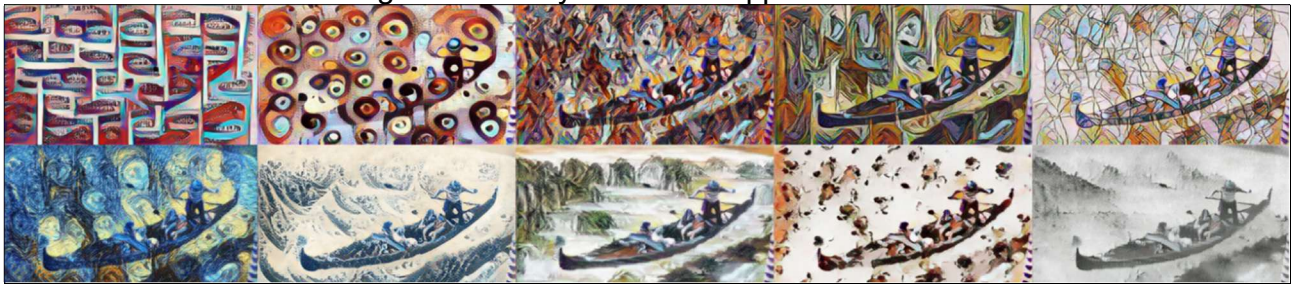
Unit 7: Streamlit for Simple AI Web Applications
- Introduction to Streamlit as a rapid web application framework.
- Installing and configuring Streamlit.
- Hands-on exercise: Building a simple AI-powered web application with Streamlit



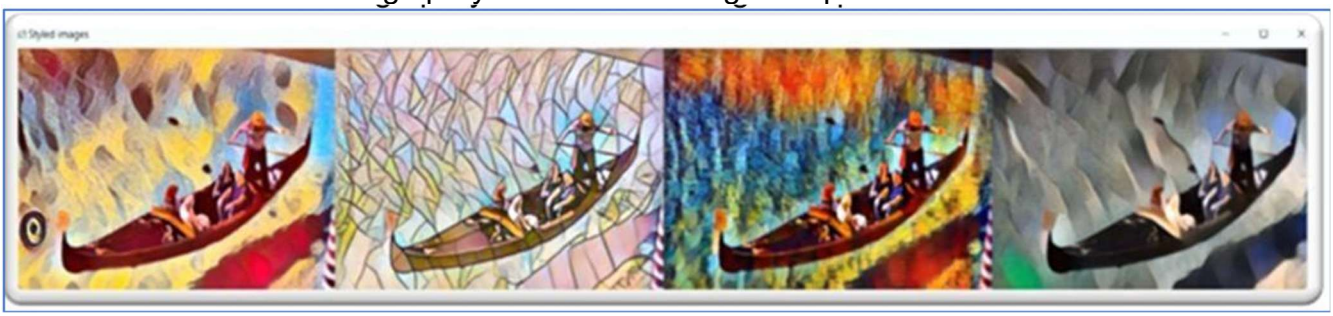
Unit 8: Local LLMs with Ollama
- Understanding the benefits and limitations of running large language models locally.
- Introduction to Ollama for managing and running open-source LLMs.
- Hands-on exercise: Setting up Ollama and running a local LLM for a specific business task.



Unit 9: Neural Style Transfer
- Introduction to Neural Style Transfer and its applications in creative AI.
- Installing TensorFlow and arbitrary image stylization tools.
- Hands-on exercise: Building a Neural Style Transfer application.



Unit 10: Introduction to PyTorch
- Introduction to PyTorch, a widely used open-source machine learning framework.
- Core concepts: tensors, automatic differentiation, neural network modules, and model training.
- Hands-on exercise: Setting up PyTorch and building a simple neural network.



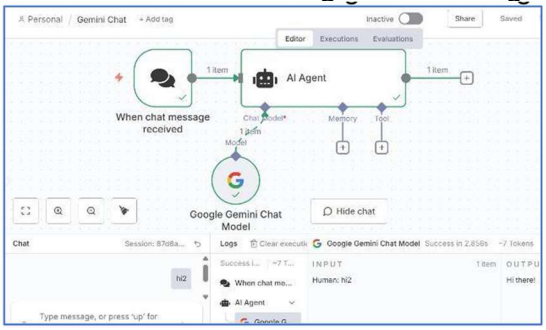
Unit 11: YOLO Object Detection with PyTorch and Ultralytics
- Introduction to YOLO as a real-time object detection framework based on deep learning.
- Installing and using Ultralytics YOLO.
- Hands-on exercise: Applying YOLO to detect objects in images and videos..

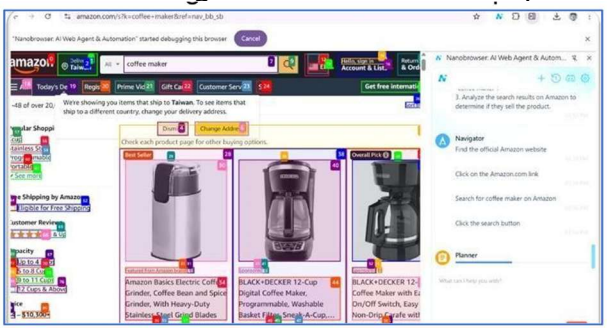


Unit 12: Using Python to Call LLM APIs
- Introduction to LLM APIs and their role in modern AI applications.
- Exploring popular LLM APIs, such as OpenAI and Google Gemini, and their Python libraries.
- Hands-on exercise: Building a Python script to call an LLM API, process the response, and apply it to a business-related task.

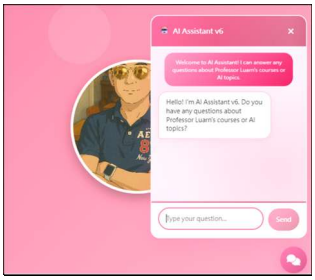


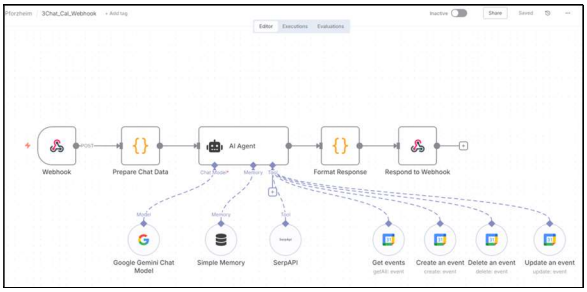
Unit 13: AI Agents, MCP, and n8n
- Introduction to AI agents and their ability to perform tasks autonomously.
- Exploring n8n as a workflow automation platform for connecting different services.
- Hands-on exercise: Creating a basic AI agent workflow using n8n to automate a simple task.





Unit 14: Advanced n8n and AI Agent Workflows
- Exploring advanced n8n features for multi-step automation.
- Building AI agent workflows that integrate multiple APIs, databases, and LLMs.
- Hands-on exercise: Developing an advanced AI agent workflow for a real-world business use case..





Unit 15: Emerging AI Coding Agents and Autonomous Development Tools
- Introduction to the latest developments in Agentic AI, with a focus on AI coding agents, autonomous development agents, tool-using agents, and persistent agents.
- Overview of emerging AI agent tools and platforms, including OpenClaw, Hermes Agent, Claude Code, OpenAI Codex, and Google Antigravity.
- Discussion of how AI coding agents can assist with code generation, debugging, refactoring, file manipulation, terminal operations, browser-based tasks, GitHub integration, and workflow automation.
- Key issues in using AI coding agents: prompt design, context management, repository access, code review, security, reliability, cost control, and human supervision.
- Hands-on exercise: Designing and testing a simple AI coding agent workflow, such as creating a personal webpage, modifying an existing project, debugging Python code, or automating a small business-related coding task.







Course Intended Learning Outcomes and their Contribution to Program Intended Learning Outcomes / Program Goals

Program Intended Learning Outcomes		Course Intended Learning Outcomes	Assessment Methods		
	After completion of the program the students will be able…	After completion of the course the students will be able…	Project Work	Participation in Class	Non-Graded Examination
			60%	30%	10%
			Individual	Individual	Collective
1	**Responsible Leadership in Organizational Contexts**				
2	**Creative Problem Solving Skills in a Complex Business Environment**				
2.2	...to analyze problems of Marketing Intelligence.	… to analyze marketing intelligence problems, through the AI application development process of this course.	X	X	X
2.3	...to develop and evaluate creative solutions to complex problems of Marketing Intelligence.	… to develop and evaluate creative solutions to complex problems through the process of learning AI application development.	X	X	X
3	**Applied Research Skills**				
4	**Interdisciplinary and agile working**				
4.2	... are able to design and manage marketing intelligence projects independently and on their own responsibility.	… to design and manage marketing intelligence projects independently and take responsibility for them.	X	X	

Teaching and Learning Approach

The course will follow a blended learning approach, combining theoretical knowledge with hands-on practical exercises. The teaching methods will include:

- Lectures: Engaging lectures to introduce and explain the theoretical concepts and principles of AI and its applications.
- Hands-on Exercises: Practical exercises using Google Colab and MediaPipe to reinforce the concepts learned and provide hands-on experience in developing AI applications.
- Project-Based Learning: Encouraging students to work on individual or group projects to apply their knowledge and skills acquired throughout the course.
- Assessments: There will be assignment and report to evaluate students' understanding and progress.
- Feedback and Support: Providing regular feedback and support to students during the project development phase, ensuring their understanding and addressing any challenges they encounter.

Literature and Course Materials

1. **Instructor-prepared lecture slides and hands-on lab materials**
   Lecture slides, guided tutorials, code snippets, business cases, and step-by-step exercises will be provided for each unit.
2. **Sample code and project templates**
   Ready-to-use sample code will be provided for computer vision, Streamlit web apps, local LLMs, LLM API calls, n8n workflows, and AI agent prototypes.
3. **MediaPipe, OpenCV, and Teachable Machine documentation**
   Official documentation and tutorials will be used to support hands-on development in visual AI, including object detection, image segmentation, face detection, pose estimation, hand landmark detection, and gesture recognition.
4. **PyTorch and Ultralytics YOLO documentation**
   Official learning materials will be used for introducing deep learning basics, tensor operations, simple neural networks, and real-time object detection with YOLO.
5. **LLM and Generative AI documentation**
   Students will refer to official documentation from OpenAI, Google Gemini, and Ollama to understand LLM APIs, local model deployment, prompt design, model interaction, and practical business applications.
6. **Agentic AI, MCP, and workflow automation materials**
   Course materials will include selected tutorials and documentation on AI agents, Model Context Protocol, n8n AI workflows, tool-using agents, memory-based agents, and autonomous task execution.
7. **AI coding agents and autonomous development tools**
   Selected materials will introduce emerging AI coding agents and development tools, including Claude Code, OpenAI Codex, Google Antigravity, OpenClaw, and Hermes Agent.
8. **Online resources and video tutorials**
   Curated online tutorials, technical blogs, documentation pages, and demonstration videos will be shared to support students' independent learning and project development.
9. Student project materials
   Students will prepare project reports, GitHub repositories, personal webpages, workflow screenshots, and prototype demonstrations as part of the course learning process..

Assessment

1. Assignments: Assignment will be given to students throughout the course, focusing onimplementing AI algorithms, developing applications, and analyzing results.

2. Quizzes: Short quizzes may be conducted to assess students' understanding of thetheoretical concepts and practical applications covered in the lectures.

3. Project Evaluation: The final projects developed by students will be assessed based oncriteria such as functionality, creativity, technical implementation, and presentation.

4. Class Participation: Active participation in class discussions, group activities, and project presentations will be considered when assessing students' engagement and contribution tothe learning process.

Schedule

	Monday Nov. 16	Tuesday Nov.17	Wednesday Nov. 18	Thursday Nov. 19	Friday Nov. 20
08:00~09:30		V			
09:45~11:15		V			
11:30~13:00					
13:45~15:15	V			V	V
15:30~17:00	V			V	V

	Monday Nov. 23	Tuesday Nov. 24	Wednesday Nov. 25	Thursday Nov. 26	Friday Nov. 27
08:00~09:30		V			
09:45~11:15		V			
11:30~13:00					
13:45~15:15	V			V	V
15:30~17:00	V			V	

Relevant Reference Websites

1. Google MediaPipe Documentation

   https://developers.google.com/mediapipe

2. Google Teachable Machine

   https://teachablemachine.withgoogle.com/

3. OpenCV Documentation

   https://docs.opencv.org/

4. Streamlit Documentation

https://docs.streamlit.io/

5. PyTorch Documentation

   https://pytorch.org/docs/stable/index.html

6. Ultralytics YOLO Documentation

   https://docs.ultralytics.com/

7. OpenAI API Documentation

   https://platform.openai.com/docs

8. Google Gemini API Documentation

   https://ai.google.dev/gemini-api/docs

9. Ollama Documentation

   https://github.com/ollama/ollama/tree/main/docs

10. n8n Documentation

    https://docs.n8n.io/

11. n8n AI Agent Node Documentation

    https://docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent/

12. Claude Code Documentation

    https://docs.anthropic.com/en/docs/claude-code

13. OpenAI Codex

    https://openai.com/codex/

14. Google Antigravity

    https://antigravity.google/

15. OpenClaw GitHub Repository

    https://github.com/openclaw/openclaw

16. Hermes Agent Documentation

    https://hermes-agent.nousresearch.com/docs/

17. GitHub Docs

    https://docs.github.com/

18. GitHub Pages Documentation

    https://docs.github.com/en/pages